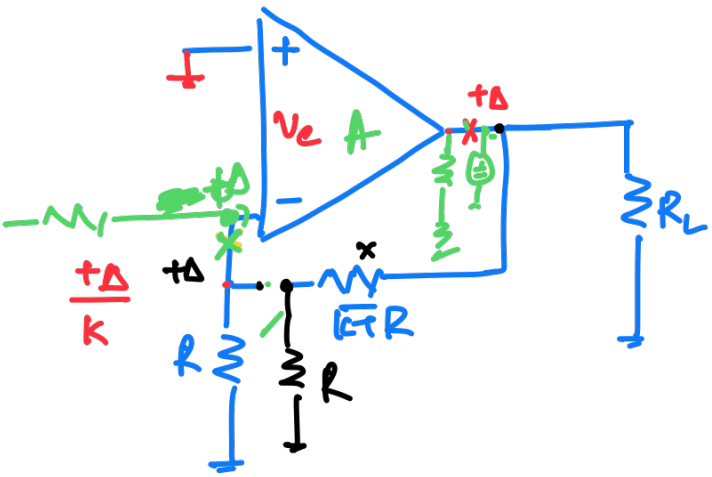
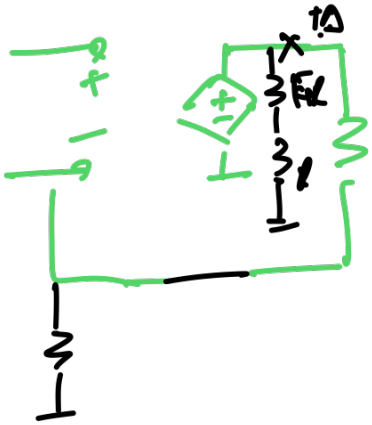
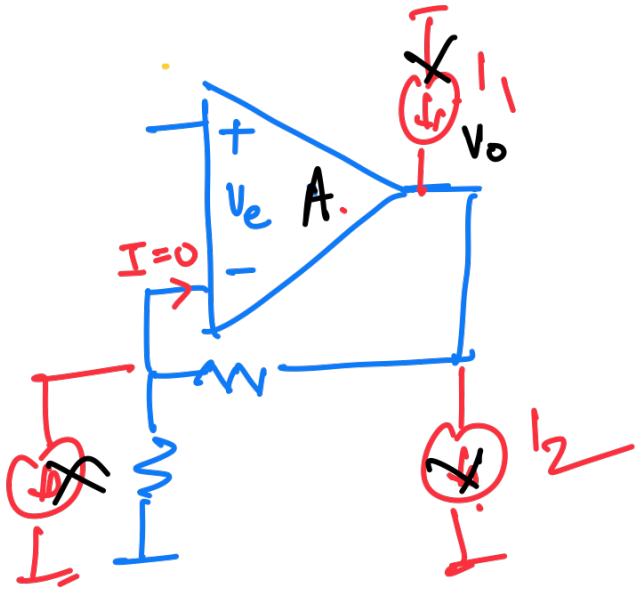
Lecture # 06





$+\Delta \longrightarrow V_e = \frac{-\Delta}{K} \xrightarrow{A} \ominus \frac{A\Delta}{K}$ Loop gain, $LG = -\frac{A}{K}$

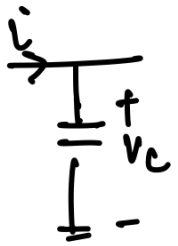
$+\Delta \longrightarrow V_e = -\Delta \xrightarrow{A} \ominus A\Delta$ Loop gain, $LG = -A\Delta$



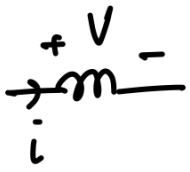
$V_o = AV_e$

$V_+ = V_-$

Integrator

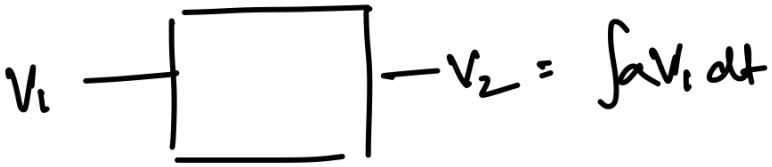


$$v_c = \frac{1}{C}\int i \cdot dt$$

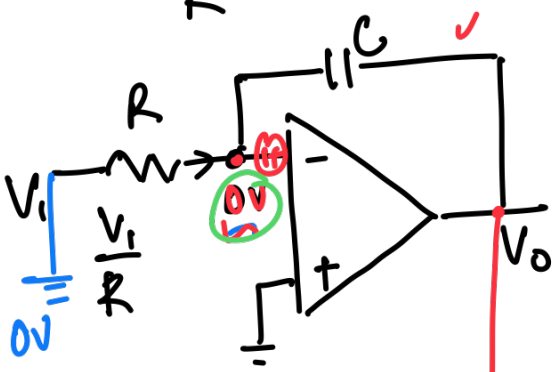


$$V = L\frac{di}{dt}$$

$$i = \int \frac{V}{L} \cdot dt$$

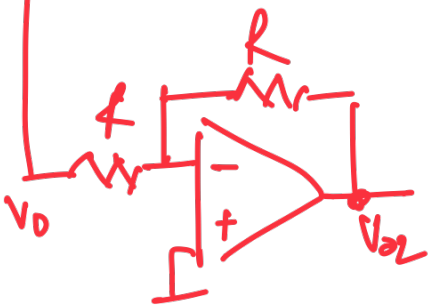


$$I = \frac{V_1}{R}$$

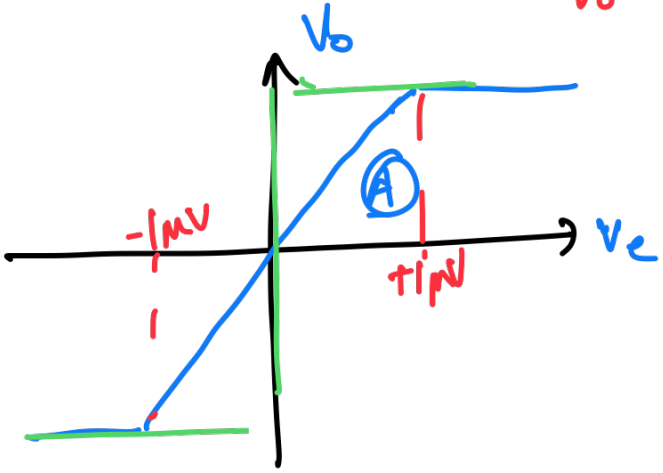


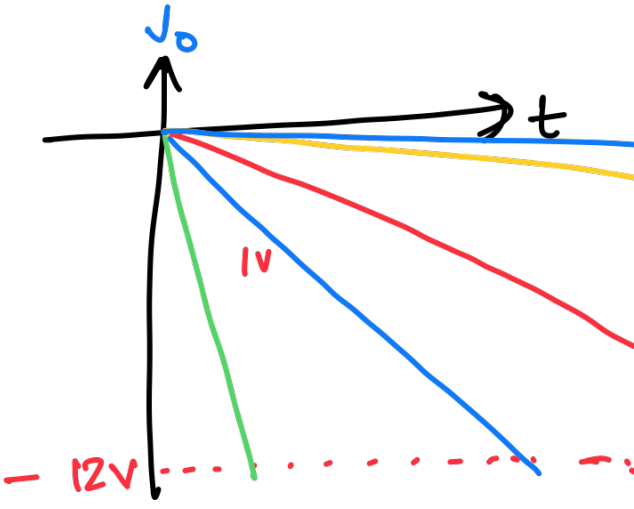
$$\frac{V_1}{R} = -C\frac{dV_0}{dt}$$

$$V_0 = -\frac{1}{RC}\int V_1 \, dt.$$

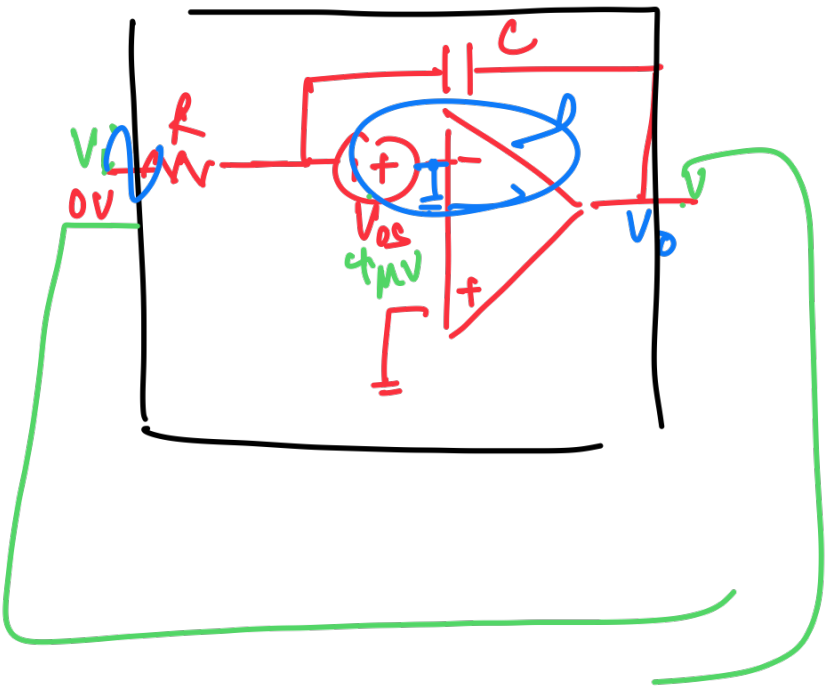


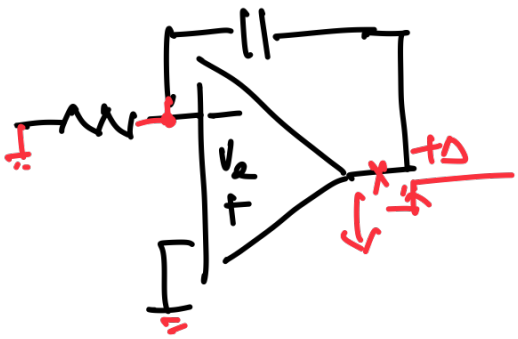
$$\frac{V_{02}}{V_0} = -1$$

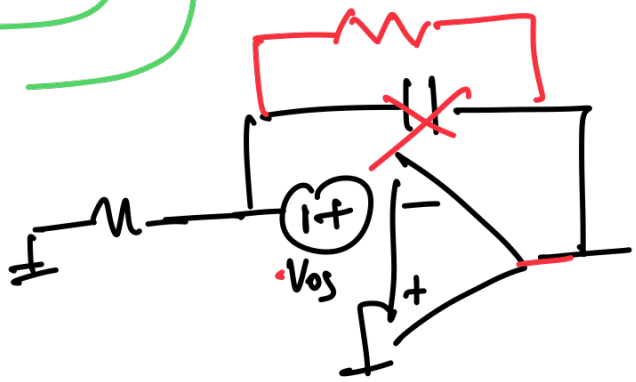




"open loop integrator"

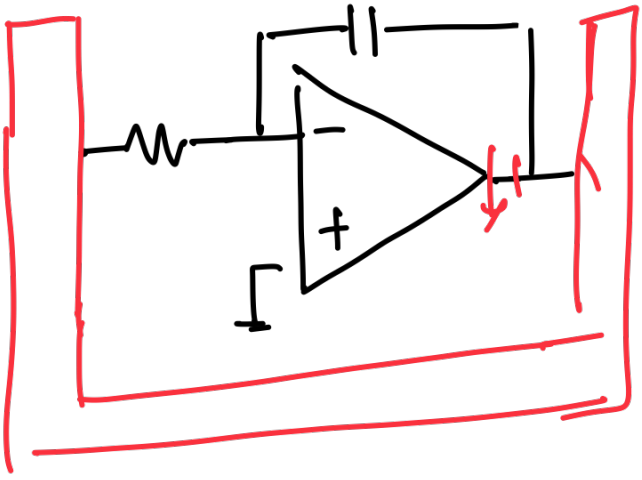




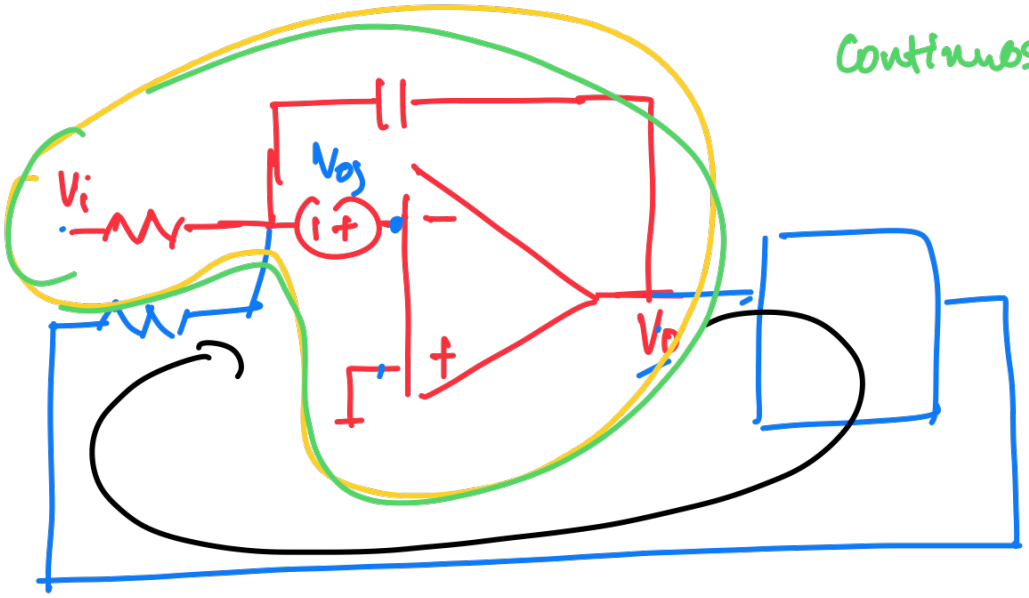


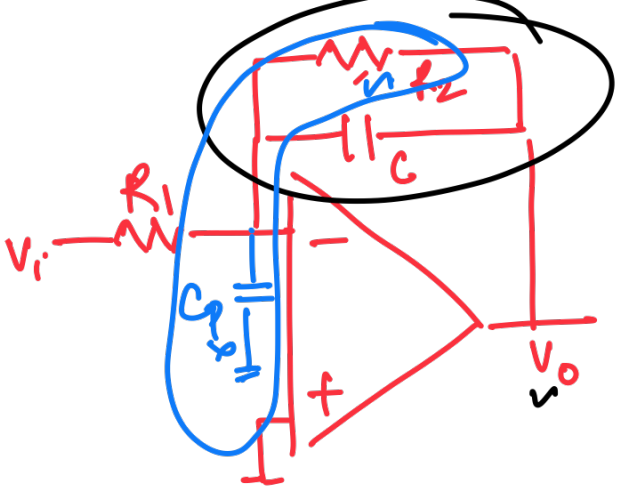
$$\frac{R}{R + 1/sC} = \frac{sRC}{1 + sRC}$$

- σ_{off}



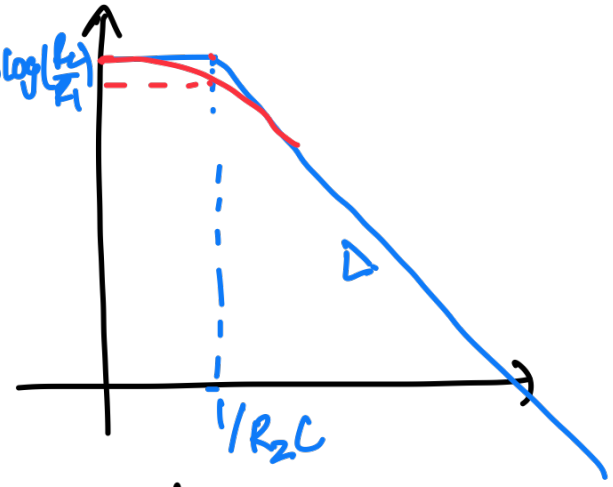
Continuous-time ΔΣ





$$\frac{V_i}{R_1} = -\frac{V_o}{R_2} - V_o sC = -\frac{V_o(1+sR_2C)}{R_2}$$

$$\frac{V_o}{V_i} = -\frac{R_2}{R_1}\frac{1}{1+sR_2C}$$



$$\left|\frac{V_o}{V_i}\right| = H(s)$$

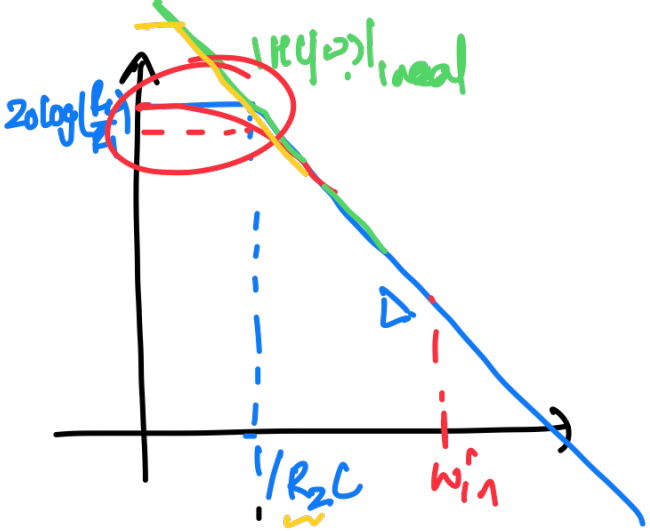
$$20\log_{10}|H(j\omega)| = 20\log_{10}\left|\frac{R_2/R_1}{\sqrt{1+\omega^2R_2^2C^2}}\right|$$

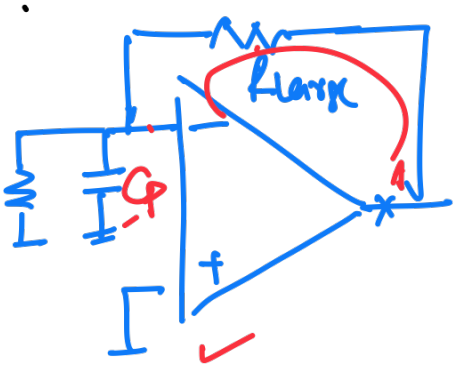
$$s = -\frac{1}{R_2C} \quad \text{(pole)}$$

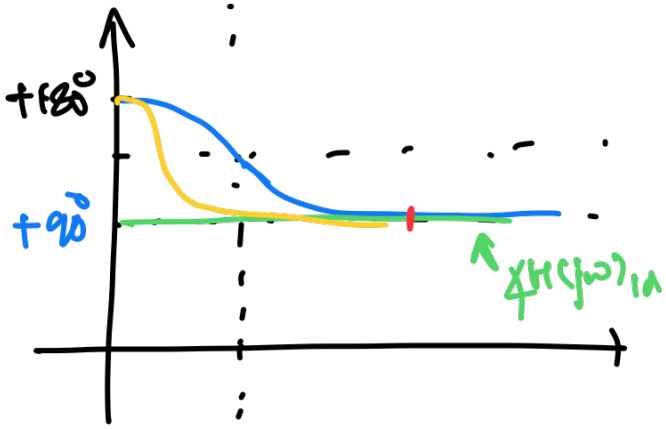
$$\omega_p = \frac{1}{R_2C} \quad \text{(pole frequency)}$$

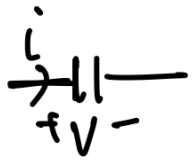
$$\angle H(j\omega) = +180^\circ - \tan^{-1}(\omega R_2C)$$

$$\left.\frac{V_o}{V_i}\right|_{\text{ideal, int}} = \frac{-1}{sR_1C} = -\frac{1/sC}{R_1}$$

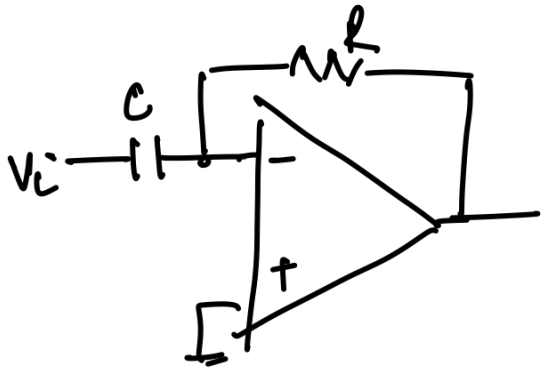




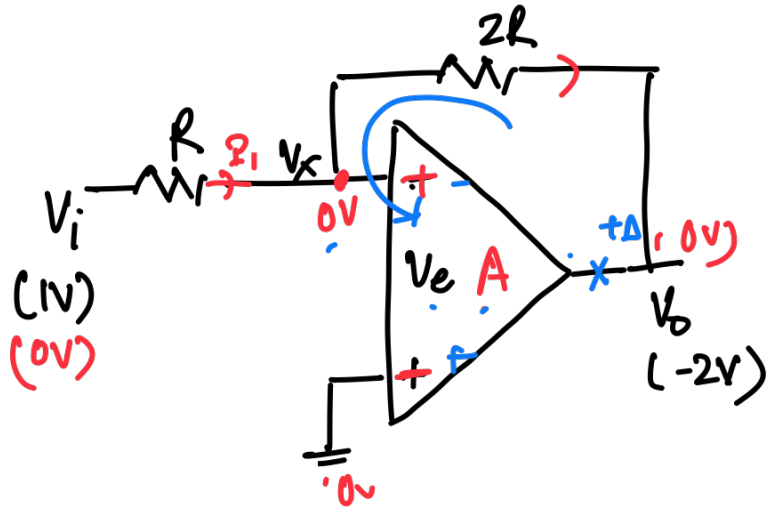




$$i = C\frac{dV}{dt}$$



$$(V_i - 0)\,sC = \frac{-V_o}{R} \Rightarrow V_o = -sRC\,V_i$$



$V_e \to 0$

$\Delta \rightarrow V_e = \frac{\Delta}{3} \rightarrow A\cdot\frac{\Delta}{3}$

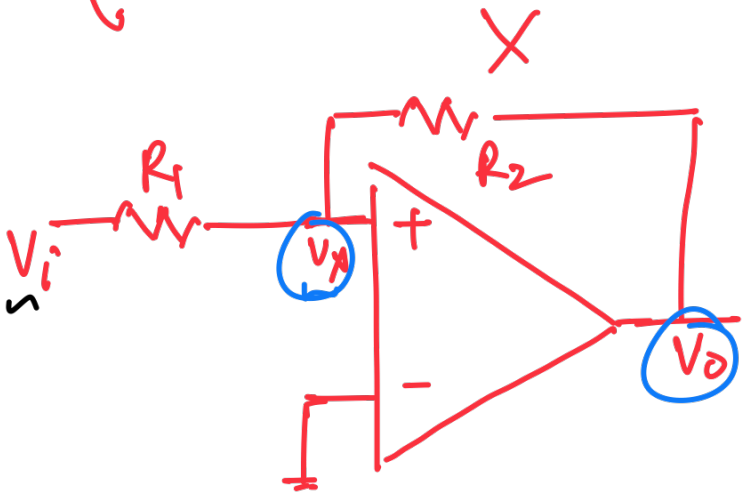
$$\frac{V_i - V_x}{R} = \frac{V_x - V_o}{2R} \Rightarrow 2V_i - 2V_x = V_x - V_o$$

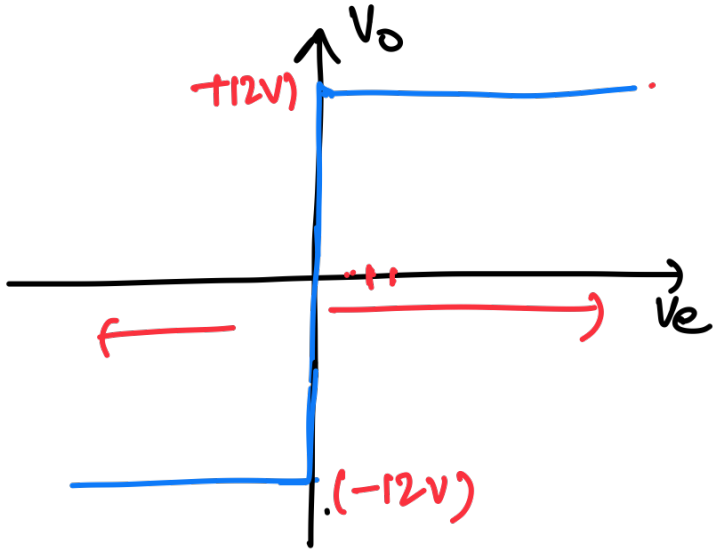
$$2V_i = 3V_x - V_o$$

$V_o = A V_x$ —

$$2V_i = \frac{3V_o}{A} - V_o$$

$$V_o = \frac{2}{\left(\frac{3}{A} - 1\right)}\Bigg|_{A\to\infty} = -2$$





$$V_o = \frac{R_2}{R_2+R_1} V_i + \frac{R_1}{R_1+R_2} V_o$$

$V_e = V_p = \frac{R_2}{R_2+R_1} V_i + \frac{R_1}{R_1+R_2} V_o$ —— $V_o = A V_e$

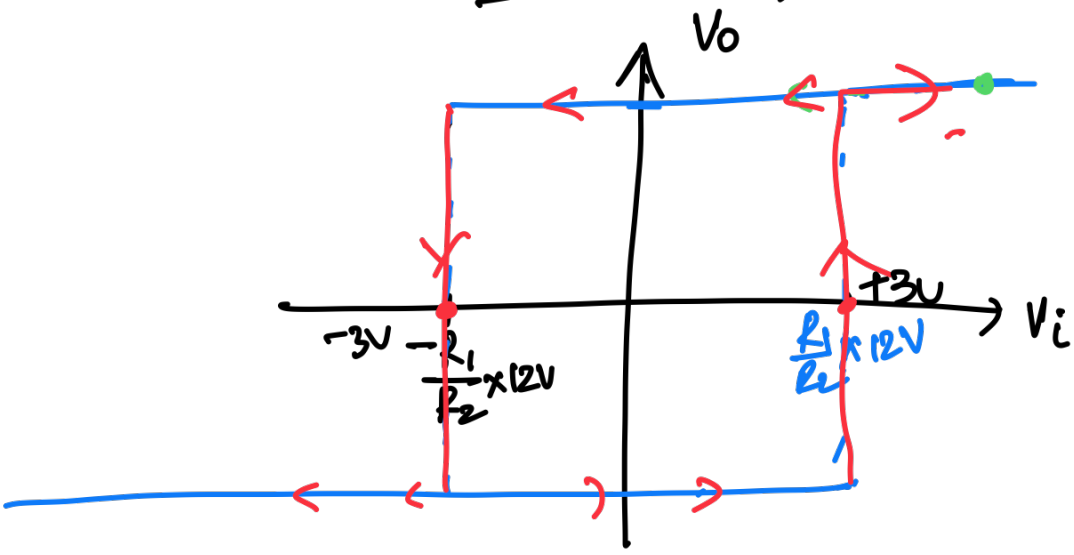
$V_e > 0 \Rightarrow \frac{R_2}{R_2+R_1} V_i + \frac{R_1}{R_1+R_2} V_o > 0$

$V_o = +12V \Rightarrow \frac{R_2}{R_1+R_2} V_i + \frac{R_1}{R_1+R_2}(+12) > 0$

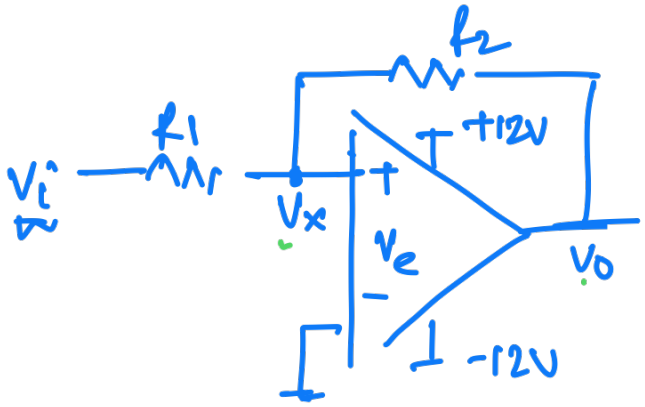
$V_i > -\frac{R_1}{R_2} \times 12V$

$V_o = -12V \Rightarrow$ ✗

$V_i > \frac{R_1}{R_2} \times 12V$



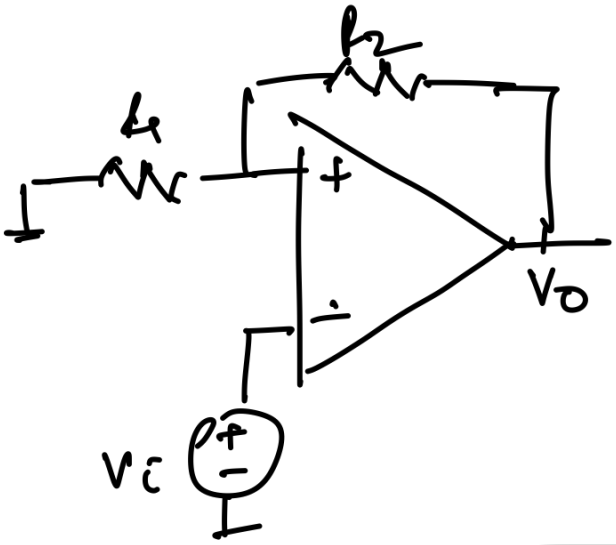
$V_e < 0$, $V_i < \frac{R_1}{R_2} \times 12V$

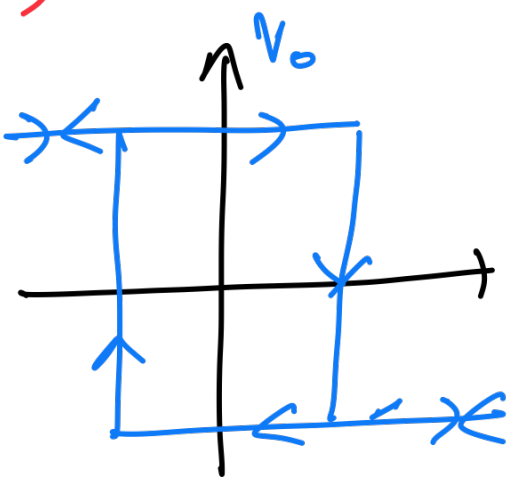


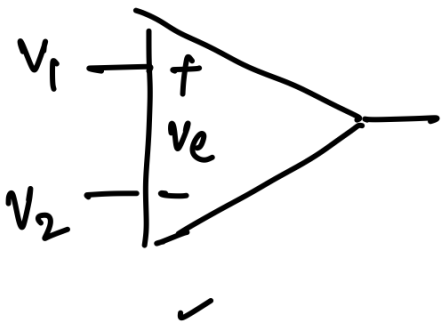
$V_i = \pm 100V$, $V_o = \pm 12V$

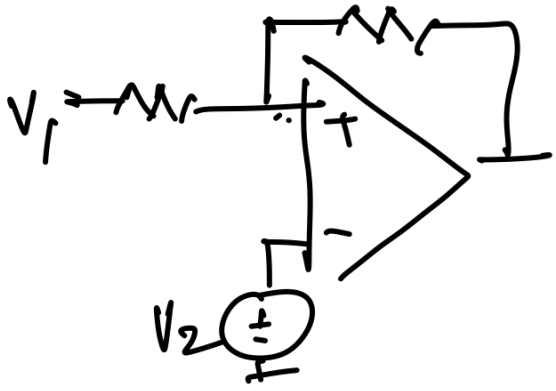
$$V_e = V_x = \frac{R_2}{R_1+R_2} V_i + \frac{R_1}{R_1+R_2} (+12)$$

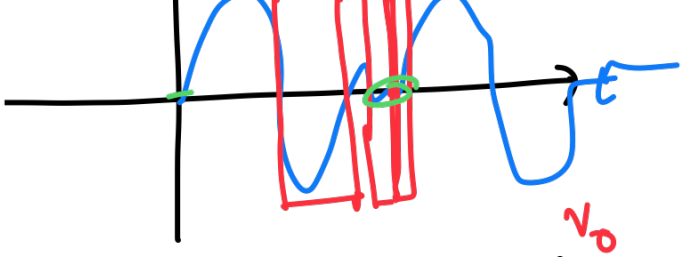
$$V_e = V_x = \frac{R_2}{R_1+R_2} V_i + \frac{R_1}{R_1+R_2} (-12V)$$

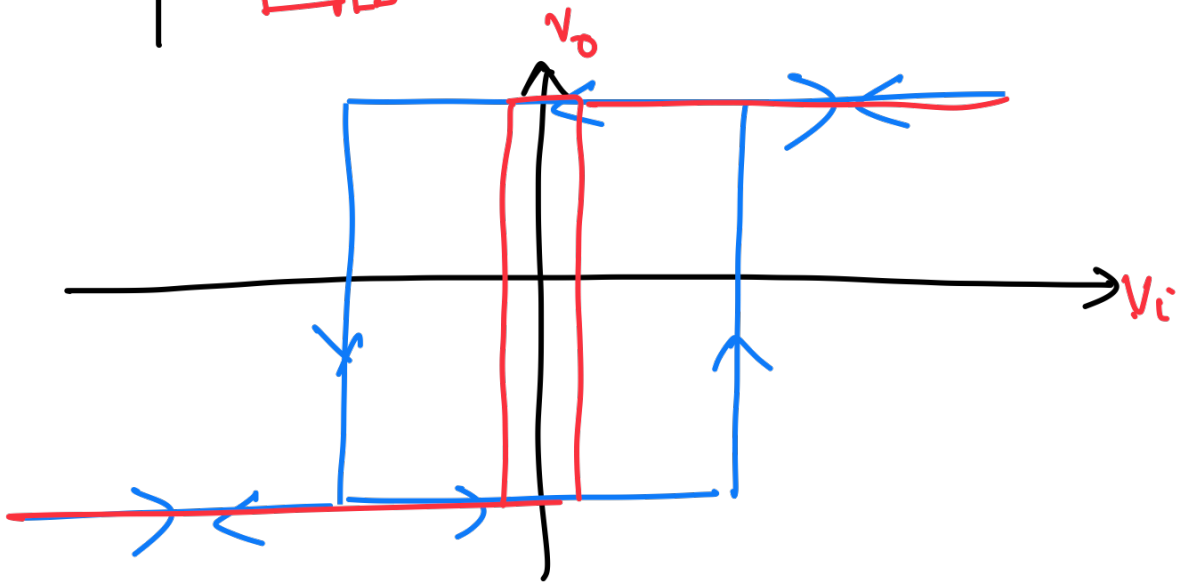




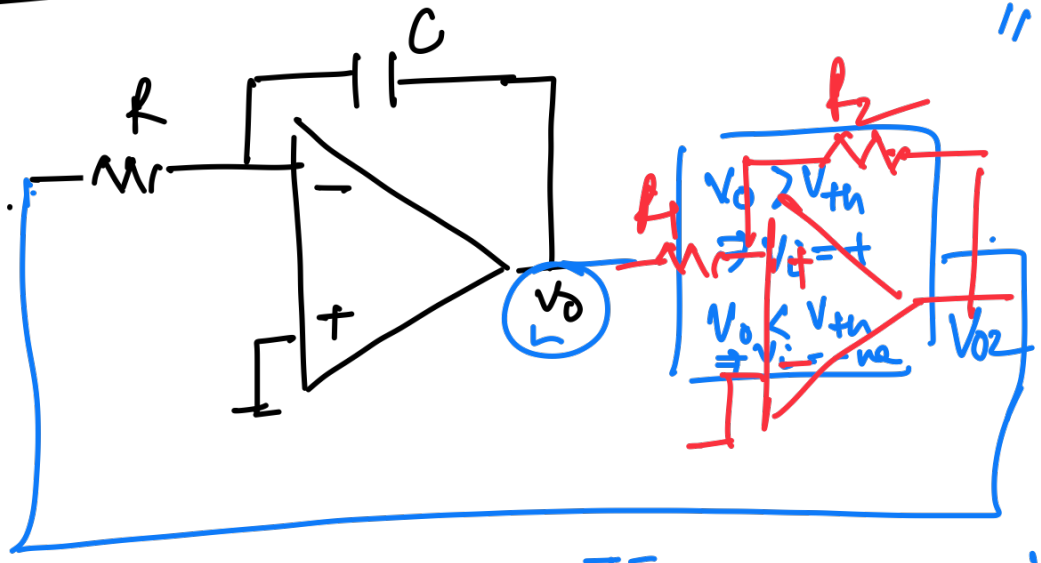


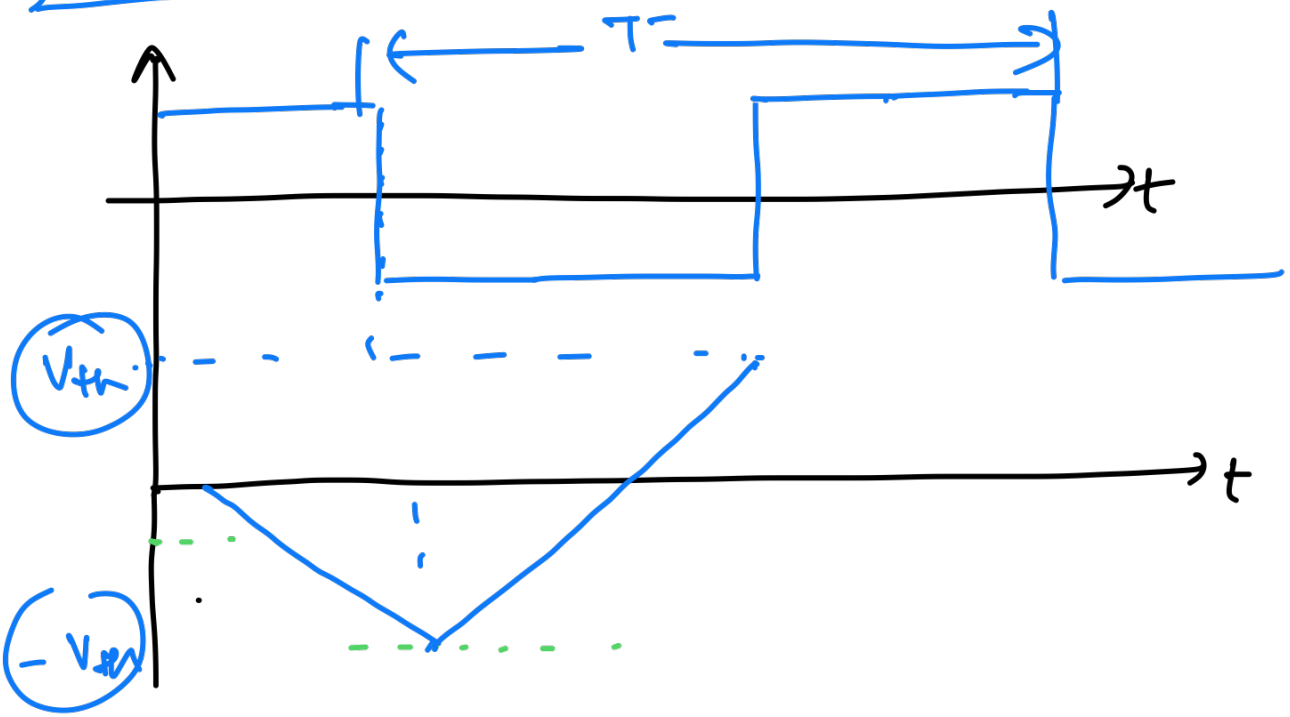






"Oscillators"





$$V_{th} = \frac{R_2}{R_1}(VDD)$$

$$\int_0^{\Delta t} \frac{VDD}{R} \cdot dt = 2V_{th}$$

$$f_{osc} = \frac{1}{T} = \frac{1}{2\Delta t}$$